

• LIVE • PAPER TRADING DESK

AegisAlpha

Autonomous Multi-Agent Quantitative Options Trading Desk

LABLAB.AI × ALPACA — AI TRADING AGENTS HACKATHON

Team Returnee • Solo Competitor

\$ live aegis-alpha-desk.vercel.app

\$ repo github.com/DeathKnell837/band-of-agents-hackathon

The Problem

Three failure modes explain why almost every autonomous options-trading agent blows up before it ever proves itself.

ERR_01

Unhedged Naked Positions

99% of LLM trading bots fail because they place unhedged naked positions with unlimited loss.



ERR_02

Zero Greeks Awareness

LLMs hallucinate position sizes and have zero Greeks awareness.



ERR_03

No Risk Separation

No existing system separates AI creativity from mathematical risk enforcement.



Our Solution

AegisAlpha is a 4-agent autonomous options desk that:



01 / STRATEGY SCOPE

Defined-Risk Spreads Only

Trades **only** Bull Call, Bear Put, and Iron Condor spreads — every position carries a hard-capped maximum loss, by construction.

BULL CALL • BEAR PUT • IRON CONDOR



02 / RISK CONTROL

Zero-Override Risk Gates

7 hardcoded, deterministic risk gates sit between every signal and the market — with **zero LLM override**.

7 GATES • 0 LLM OVERRIDES



03 / EXECUTION MODEL

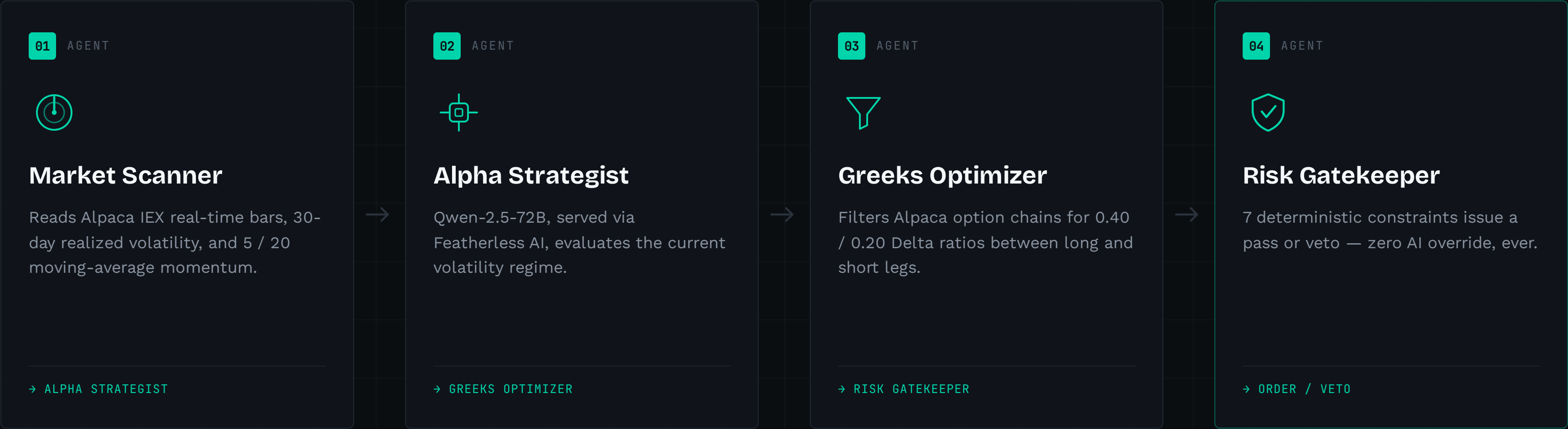
AI Strategy, Math Execution

Qwen-2.5-72B proposes the trade thesis; pure deterministic math validates and executes it.

STRATEGY: LLM • EXECUTION: MATH

Multi-Agent Architecture

Market data becomes a risk-approved order through a four-agent pipeline — each agent has exactly one job, in sequence.



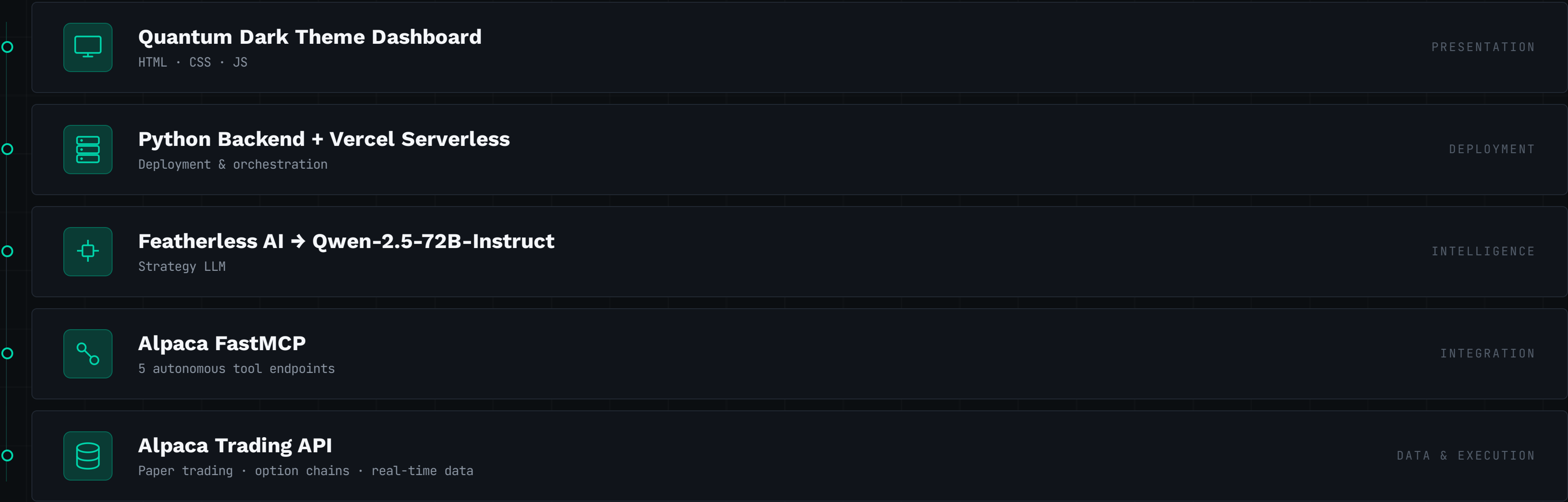
The 7 Risk Guardrails

Every proposed trade passes through all seven gates before it can reach the market. Any single failure blocks, resizes, or freezes the order — no exceptions.

GATE	RULE	ACTION
1	Defined-Risk Spreads Only	HARD BLOCK
2	Max 2% Equity Risk \$2,000 max per trade	VETO / RESIZE
3	20% Portfolio Options Cap \$20,000 max	HARD BLOCK
4	Bid-Ask Spread < 15%	SLIPPAGE FILTER
5	5–45 DTE Window	GAMMA PIN GUARD
6	-3% Daily Drawdown circuit breaker	FREEZE
7	Dynamic Position Downscaling	AUTO SCALE

Tech Stack

Five layers, from live market data at the base to the dashboard operators watch on top.



FastMCP Integration

Five MCP tool endpoints expose AegisAlpha's entire trading loop to any AI IDE — no custom integration required.

aegisalpha_mcp_server.py

```
01 mcp.tool → get_account_telemetry ()
02 mcp.tool → scan_and_propose_spread ()
03 mcp.tool → validate_risk_gate ()
04 mcp.tool → execute_options_order ()
05 mcp.tool → emergency_kill_switch ()
```

COMPATIBLE WITH

■ Cursor

■ Claude Desktop

■ Antigravity

Live Dashboard Demo

The Quantum dark-theme dashboard operators use to watch, override, and kill the desk in real time.

aegis-alpha-desk.vercel.app

UI PREVIEW • PAPER ACCOUNT

PORTFOLIO EQUITY

\$102,481

P&L (DAY)

+\$1,204.50

BUYING POWER

\$48,900

⏻ EMERGENCY KILL SWITCH

OPTIONS PAYOFF CURVE

MAX PROFIT

MAX LOSS

DECISION FEED

IRON CONDOR • SPY

APPROVED

BULL CALL • QQQ

VETOED - GATE 2

BEAR PUT • AAPL

APPROVED

GREEKS MATRIX

Δ	Γ	Θ	V	IV
0.38	0.04	-12.6	8.2	24.3%
DELTA	GAMMA	THETA	VEGA	IMPL VOL

AGENT PIPELINE

• Market Scanner

ACTIVE

• Alpha Strategist

ACTIVE

• Greeks Optimizer

ACTIVE

• Risk Gatekeeper

ACTIVE

🔍 Search option chain — SPY, QQQ, AAPL...

LABLAB.AI × ALPACA — AI TRADING AGENTS HACKATHON

What Makes Us Different

Same market, same options chains — a fundamentally different relationship between the AI and the risk.

FEATURE	OTHER BOTS	AEGISALPHA
Risk Control	✗ LLM decides	✓ 7 hardcoded math gates
Strategy Type	✗ Naked / directional	✓ Defined-risk spreads only
Max Loss	✗ Unlimited	✓ \$2,000 per trade (2%)
Architecture	✗ Single agent	✓ 4-agent dialectic pipeline
Deployment	✗ Local only	✓ Vercel + FastMCP

• END OF DECK

Links & Contact

AEGISALPHA — AUTONOMOUS MULTI-AGENT QUANTITATIVE OPTIONS TRADING DESK

```
$ live      aegis-alpha-desk.vercel.app
$ slides    aegis-alpha-desk.vercel.app/slides.html
$ repo      github.com/DeathKnell837/band-of-agents-hackathon

$ team      Returnee · Alpaca Account PA3PL5AZ85K6
```